ASAD HUSSAIN

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PROFESSIONAL SUMMARY

Final-year Electrical Engineering student at NUST working across two tracks: applied machine learning and embedded/RF systems engineering. On the AI side, building content-scoring ML pipelines at FlyRank AI and trained in deep learning, NLP, LLMs, and RAG through AtomCamp's AI Bootcamp. On the hardware side, designed DroneGuard — a two-board RF detection and jamming system spanning four frequency bands, GNU Radio signal classification, and real-time MATLAB visualization — alongside embedded firmware and IoT telemetry work from prior internships. Comfortable moving between training models and designing the circuits and firmware that feed them real-world data.

WORK EXPERIENCE

Machine Learning Intern | *FlyRank AI (Chicago, Illinois, USA — remote)* Aug 2026 – Present

- **Content Opportunity Scoring:** Working on Lane 2 of FlyRank's ML pipeline, building models that flag content pages with strong search visibility but low click-through rates for prioritized refresh.
- **Data Contracts & Baselines:** Built a data contract notebook defining schema and validation rules for the scoring pipeline, and a baseline action-score notebook establishing reference performance before model iteration.
- **Structured ML Workflow:** Worked through a staged ML track (data contracts, EDA, baseline modeling) applied to real SEO/content datasets rather than toy problems.
- **Case Study Documentation:** Authored portfolio case-study write-ups translating model outputs into content-refresh recommendations for stakeholders.

Intern — IT & Technical Departments | *National Electric Power Regulatory Authority (NEPRA), Islamabad* Jul – Aug 2026

- **Licensing & Tariff Framework:** Studied NEPRA's end-to-end regulatory flow — license → tariff → sale — across Generation, Transmission, Distribution, and Supply licenses, including the three generation tariff mechanisms (cost-plus, upfront, competitive bidding).
- **Document Intelligence Portal:** Contributed to a web-based portal for centralizing and full-text searching NEPRA's regulatory filings, covering direct upload, web crawling, and indexed search across PDF/DOCX/XLSX/PPTX formats.
- **Power Systems Study:** Studied Advanced Metering Infrastructure (AMI) architecture and power transformer construction, ratings, losses, and protection/testing practices in the Technical Department.
- **Regulatory Analysis:** Reviewed NEPRA's State of the Industry Report 2025 (generation, transmission, distribution) and compiled a full internship report and technical reference guide on findings.

Engineering Intern — | *Research & Indigenous Development Centre (RDC) Heavy Industries Taxila (HIT)* Jun – Aug 2025

- Integrated and wired core hardware modules including the LilyGO T-SIM7670E (ESP32 core + LTE modem), LilyGO T3-S3 GPS module, and RDM6300 RFID reader with a TP4056 LiPo charge controller.
- Programmed low-level control logic and state machines in C/C++ for 1Hz GPS data capture, RFID card authentication, and buzzer-based audio feedback.
- Implemented MQTT messaging over cellular GPRS with QoS Level 1, enabling reliable real-time telemetry with 50ms server-side callback processing.
- Built an asynchronous Python Flask backend using multithreading to process analytics workloads without blocking incoming MQTT streams.

FINAL YEAR PROJECT

DroneGuard — Anti-Drone Surveillance System 2025 – 2026

SDR · RTL-SDR · GNU Radio · Multi-Band RF Receivers · Raspberry Pi 4 · STM32H743 · MATLAB · Python

- Designed a two-board anti-drone system: an RF detection front-end covering four bands (433 MHz, 915 MHz, 240–930 MHz sweep, 2.4 GHz) using CC1101, Si4432, and NRF24L01+PA+LNA transceivers, and a jamming board built around an ADF4351 PLL synthesiser.
- Integrated an SPF5189Z low-noise amplifier (+18.7 dB gain, 0.6 dB noise figure) ahead of the receiver chain, extending detection range from roughly 30 m to 150–200 m for typical consumer drone control links.
- Built a real-time signal-classification pipeline on a Raspberry Pi 4 using an RTL-SDR dongle and GNU Radio, identifying FHSS, OFDM, and FSK drone protocols (DJI OcuSync, FrSky, ArduPilot/SiK) from live IQ samples.
- Developed a MATLAB front-end app displaying live PSD spectrum, spectrogram waterfall, and GPS/telemetry data over a UDP link from the Raspberry Pi, alongside YOLOv8-based visual confirmation.

- Authored a full component-level technical reference (RF chain design, power budgeting, PLL calibration) and costed the system at roughly a 98% reduction versus commercial anti-drone platforms.

EDUCATION

National University of Sciences and Technology (NUST) 2023 – 2027

B.E. Electrical Engineering

Relevant coursework: Analog Electronics · DSP · Control Systems · Communication Systems · Digital Logic Design · Embedded Systems · Signals & Systems · Instrumentation · OOP · Microprocessor Systems

AI/ML TRAINING — ATOMCAMP

atomcamp is a Pakistan-founded AI & data science training academy that has trained 6,000+ professionals worldwide and expanded internationally as atomcamp Arabia — delivering AI bootcamps across Saudi Arabia in partnership with the Kingdom's Ministry of Communications and Information Technology (MCIT), alongside operations in the UK and Malaysia. Completed the AI Bootcamp, covering:

- **Machine Learning & Deep Learning:** Supervised/unsupervised learning, regression, SVM, decision trees, random forests, clustering, CNNs, transfer learning, and YOLO-based object detection.
- **NLP & LLMs:** Tokenization, embeddings (Word2Vec, GloVe), transformer architecture, and fine-tuning large language models.
- **Generative AI & RAG:** Multimodal generative AI, LangChain-based agent development, and retrieval-augmented generation pipelines.
- **No-Code AI Agents (n8n):** Built drag-and-drop automation agents connecting APIs, webhooks, and data sources for research/reporting workflows.
- **MLOps & Deployment:** Git, Docker, FastAPI, Weights & Biases experiment tracking, CI/CD with GitHub Actions, and deployment on Google Cloud (Cloud Run/App Engine).

ACADEMIC PROJECTS

40 + hands-on projects across core EE disciplines. Key highlights: Butterworth Active LPF (op-amp) · Multi-Stage Audio Amplifier · EEG Cleaning MATLAB GUI · PID DC Motor Controller (Arduino) · 5-Band Graphic Equaliser (Simulink) · 4-Bit ALU · FM Receiver (TDA7000) · Multi-Range Voltmeter · OOP Shopping Cart (C++) · Water Level Indicator (Proteus)

TECHNICAL SKILLS

Programming: Python · C++ / Embedded C · JavaScript / HTML / CSS · SQL · MATLAB / Simulink · Flask

Hardware & RF: SDR · Antenna Design & Optimisation · STM32 · AVR · PIC · Arduino · Raspberry Pi · PCB Design (Proteus) · SPICE

ML / AI: Scikit-learn · TensorFlow · PyTorch · LangChain · RAG · LLM fine-tuning · n8n · FastAPI · Docker · W&B

Software & Tools: SQL Server · REST APIs · Git · AutoCAD · Proteus · Google Cloud

Signal Processing: FFT / Spectrogram Features · RF Signal Classification · MATLAB ML Toolbox · Dataset Labelling

CERTIFICATIONS & ACTIVITIES

AtomCamp AI Bootcamp — Machine Learning, Deep Learning, NLP, Generative AI/RAG, n8n ·

MATLAB Fundamentals — MathWorks ·

PCB Design — Proteus & SPICE ·

Technical Lead, NUST Robotics Society · Active Member, NUST Engineering Society

PORTFOLIO

asad-portfolio-7unj.onrender.com — Full-stack portfolio site showcasing projects across embedded systems, RF/SDR, and ML/AI, with dynamic API-driven project pages, certificate lightbox, and contact form. Deployed on Render.